Will this workout be hard for this person?

Machine Learning Hackathon 2026 · 30 October · 13:00–19:30

Think first, then prompt.

Why your data matters

Last year's lesson: Python code alone no longer shows what you learned, because it can be generated in minutes. We need to look at the process: how you plan, and how well you understand the code you generate.

1 What we would study

Your prompts, notes and paper sheets, to see what really helps you learn, and what does not.

2 An open question

Nobody knows yet how to teach and assess knowledge in the age of AI.

3 Who benefits

More than giving data: you help improve our teaching and help the students of the coming years.



Goal: about 150 participants, nearly 9 in 10 students across both classes. The phenomenon is complex and needs data: every answer counts.

Still your choice: saying no changes nothing to your grade.

The EPIAGEN research team



Québec, Canada

ÉTS Montréal · Université TÉLUQ

France

ECE Paris · Sorbonne Université · Paris 8 · Sorbonne Paris Nord · Université de Lorraine

AIED 2027, Australia

June 2027: where we hope to present the results

If the paper is accepted at AIED 2027, it will thank ECE's class of 2028 for its help. My collaborators and I thank you.

Research participation

1. Voluntary. Saying no changes nothing to your grade.
2. The consent form goes into a sealed envelope: your teacher only learns who agreed after grades are published.
3. Everything is pseudonymised with your participant code.
4. You can withdraw until 31 January 2027.

Your afternoon at a glance

13:00
Welcome, teams

13:10
Pre-test

13:35
GLAT

13:50
Kick-off + #START

14:05
Team plan (no AI)

14:20
Sprint 1

15:50
Gate 1 · break

16:20
Hint 1 · Sprint 2

17:35
Gate 2

18:10
Kaggle closes · hand-in

18:20
Explanation sheet

18:40
Post-test

19:05
Questionnaire

19:20
Results

Individual, on paper, devices closed

Team work, AI allowed

Deadline / checkpoint

The challenge

Real workouts from users of a fitness tracking app

user_id	sport	gender
start time	duration	distance
speed	max speed	climbing

Predict how hard each workout was for that person

easy **moderate** **hard**

Measured by the app's sensors, separately for every person. The three classes are about equally frequent. What "hard" means is part of the puzzle.

The Kaggle Pipeline



Half of the test set gives the public leaderboard you see today. The other half stays hidden and decides the final ranking, so overfitting the public score does not pay.

Join the Kaggle competition

30 October · international class

1. Every student scans the code or opens the link with their own Kaggle account.
2. Accept the competition rules.
3. Do not submit yet: form your team first (next slide).

`https://www.kaggle.com/t/1f42e0a7adef421e840c1e66a2d6daad`

Scan to join

Form your Kaggle team first

1

Leader

Joins the competition, opens the Team tab, invites every member

2

Members

Join the competition and accept the merge

3

Check

Everyone sees the same team name and all members

4

Only then

The first CSV is uploaded

!

Nobody uploads anything before the team is complete, not even sample_submission.csv. If you submit alone first, Kaggle's quota rule can block the merge later.

Team names and accounts

1

Official name only

Name the Kaggle team Team_13, Team_14 … Team_28. Solo accounts and unknown names on the private leaderboard are ignored for grading.

2

One account per student

All 6 or 7 members inside the Kaggle team. No accounts kept "in reserve" to test models on the side.

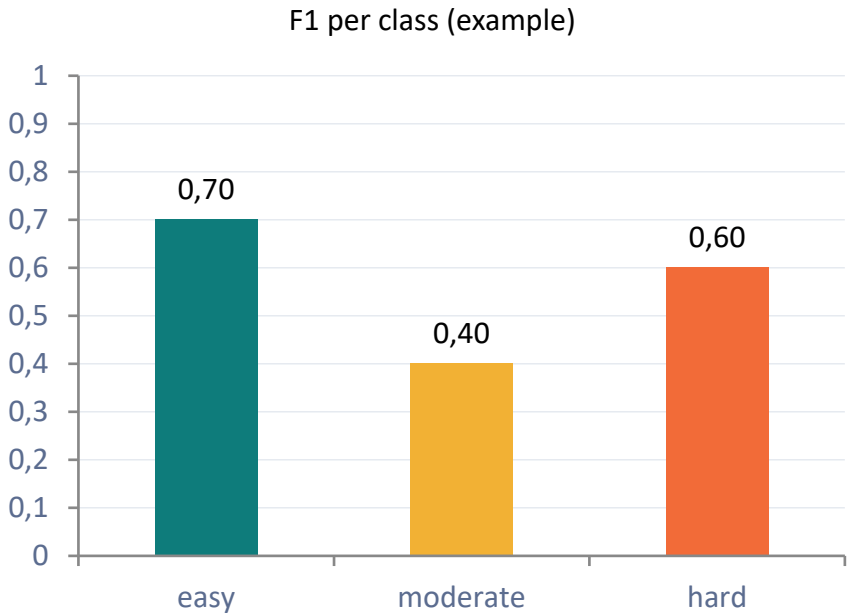
3

Checked at 14:20

The teacher opens the Team tab and compares member counts with the official lists. Missing members merge immediately.



The metric: macro-F1



macro-F1 = (0.70 + 0.40 + 0.60) / 3 = 0.57

1. F1 is computed for each class, then averaged.
2. Every class counts the same: ignoring "moderate" costs you a third of the score.
3. A model that always says "easy" gets about 0.17.

The rules

Allowed

- Any scikit-learn model that is not a neural network: course models, k-NN, SVM, naive Bayes, extra trees, HistGradientBoosting, voting, stacking
- pandas, NumPy, matplotlib
- Generative AI assistants, in a new conversation

Not allowed

- Deep learning, including MLPClassifier
- External data
- Sharing code between teams
- More than 15 Kaggle submissions per team

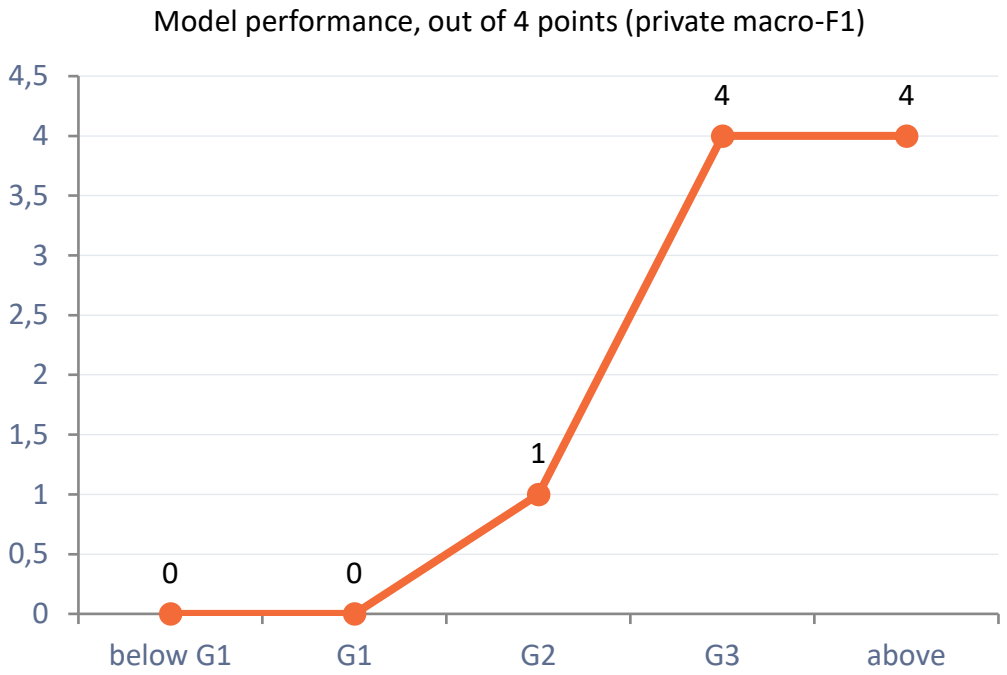
You submit it, you explain it: every member must explain the final model on paper.

Gates: a floor, not a finish line

macro-F1 ↑

G3 = ?	**Gate 3 · 17:35**	Revealed at 17:35: the teacher's reference. Reach it for the full 4 performance points
G2 = ?	**Gate 2 · 17:35**	Revealed at 17:35. Public score ≥ G2 and a confusion matrix with one sentence of error analysis in LOG.md → +1 point
G1 = ?	**Gate 1 · 15:50**	Revealed at 15:50. Public score ≥ G1 → +1 point

Where the performance points are



below G1	0
G1 → G2	0 → 1
G2 → G3	1 → 4
≥ G3	4

A quick default random forest lands around G2: 1 point out of 4. The other 3 points are for teams that understand the problem.

Your grade, out of 20

11 points individual | **9 points team**



Bonuses: top 3 teams +0.5 · best idea in LOG.md +0.5

Peer ratings can reduce the team points of a member who did not contribute. AI conversations are required but never graded on content.

Hints

1

Hint 1

16:20

Free, for everyone

2

Hint 2

after 16:20

Costs 0.5 performance point

3

Hint 3

after 16:20

Costs 1 performance point

Only the team leader can buy a hint, from the teacher. Teams that find the idea on their own keep all their points.

LOG.md: one line per submission

Time	What changed	Why we expect it to help	Validation	Public
14:55	Logistic regression, one-hot sport	Baseline with all columns	0.39	0.40
15:30	Random forest, max_depth=12	Non-linear effects of speed	0.50	0.51
16:45	…	…	…	…

Write the hypothesis before you look at the score. At least 3 model families compared on validation. Gate 2 adds a confusion matrix and one sentence on which classes are confused.

Your AI conversations



Open a new conversation in every assistant you use, and type these markers when announced.

Share the link(s) after #END, on Boostcamp.

Required for everyone; never graded on content. Remove personal information.

How to share a conversation

ChatGPT	Share (top right) → Create link → Copy
Claude	Share (top right) → public link → Copy
Gemini	Share → Share conversation → Create public link
Copilot / Le Chat	Share icon → Copy link
GitHub Copilot in VS Code	Ctrl+Shift+P → Chat: Export Chat… → upload the .json
Anything else	Copy the whole chat into a .txt or PDF and upload it

A link is a snapshot: create it after #END, or the end is missing.

Paper moments

1	**Pre-test**	13:10 · 25 min
2	**GLAT**	13:35 · 15 min
3	**Team plan**	14:05 · 15 min
4	**Explanation sheet**	18:20 · 20 min
5	**Post-test**	18:40 · 25 min

- Devices closed, phones in your bag.
- Your participant code is pre-printed: never write your name.
- Your teammates get different test versions, and your post-test version is one nobody in your team saw before.
- Do not photograph or share the questions, including with the other class.

18:10: hand-in checklist

- [ ] Kaggle: the team's final submission is selected
- [ ] Boostcamp "Code": team notebook + LOG.md (each member uploads)
- [ ] Boostcamp "AI conversations": all links or files + declaration
- [ ] **Close your laptop: explanation sheet, post-test, questionnaire**



Think first, then prompt

Good luck!